

Database Final Project

Cybersecurity Club Membership & Budget Tool

[Github repository](#)

Marissa Ratschki

ABSTRACT

This project implements a relational database for managing Cybersecurity Club members, events, attendance, and budget transactions. The database addresses the problem of club information being scattered across spreadsheets, group chats, and informal notes. It provides one organized system that executive board members could use to track who is in the club, which events were held, who attended each event, and where club money was allocated.

SCHEMA

Tables:

- Members: stores member information such as name, email, major, class year, and club role.
- Events: stores club event information such as event name, type, date, location, and description.
- Attendance: connects members to events they attended.
- Budget_transactions: stores income and expense records for the club.

Keys and Relationships:

- Members.member_id is the primary key for members.
- Events.event_id is the primary key for events.
- Attendance uses a composite primary key of member_id and event_id.
- Attendance.member_id references members.member_id.
- Attendance.event_id references events.event_id.
- Budget_transactions.transaction_id is the primary key.
- Budget_transactions.event_id is a nullable foreign key that refers to events.event_id.

QUERY LIBRARY

Query 1, basic Filtering:

```
SELECT
  member_id,
  first_name,
  last_name,
  major,
  class_year,
  member_role
FROM members
WHERE member_role = 'board'
ORDER BY last_name, first_name;
```

OUTPUT:

1	"Autumn"	"Eagan"	"Cybersecurity"	"Senior"	"board"
4	"Megan"	"Scott"	"Information Systems"	"Junior"	"board"

Query 2, joining attendance events, and members:

```
SELECT
  e.event_name,
  e.event_date,
  m.first_name || ' ' || m.last_name AS member_name,
  a.attended_at
FROM attendance a
JOIN members m
  ON a.member_id = m.member_id
JOIN events e
  ON a.event_id = e.event_id
WHERE e.event_name = 'HackTheBox Workshop'
ORDER BY member_name;
```

OUTPUT:

"HackTheBox Workshop"	"2026-03-03"	"Ana June"	"2026-03-03 18:40:00"
"HackTheBox Workshop"	"2026-03-03"	"Autumn Eagan"	"2026-03-03 18:35:00"
"HackTheBox Workshop"	"2026-03-03"	"John Shelby"	"2026-03-03 18:42:00"
"HackTheBox Workshop"	"2026-03-03"	"Tyler Langley"	"2026-03-03 18:37:00"

Query 3, event related expenses:

```
SELECT
  bt.transaction_date,
  e.event_name,
  bt.category,
  bt.amount,
  bt.description
FROM budget_transactions bt
JOIN events e
  ON bt.event_id = e.event_id
WHERE bt.transaction_type = 'expense'
ORDER BY bt.transaction_date;
```

OUTPUT:

"2026-01-26"	"Spring CTF Practice"	"competition"	100.00	"CTF practice platform fee"
"2026-03-03"	"HackTheBox Workshop"	"food" 46.25		"Pizza for HackTheBox workshop"
"2026-03-15"	"Alumni Cybersecurity Panel"	"food" 25.00		"Snacks for alumni panel"
"2026-04-12"	"Internetworking Review"	"supplies" 35.00		"Whiteboard markers for review session"

Query 4, attendance (per event):

```
SELECT
  e.event_id,
  e.event_name,
  e.event_date,
  COUNT(a.member_id) AS attendance_count
FROM events e
LEFT JOIN attendance a
  ON e.event_id = a.event_id
GROUP BY e.event_id, e.event_name, e.event_date
ORDER BY e.event_date;
```

OUTPUT:

3	"Spring CTF Practice"	"2026-01-26"	2
1	"HackTheBox Workshop"	"2026-03-03"	4
2	"Alumni Cybersecurity Panel"	"2026-03-15"	4
4	"Internetworking Review"	"2026-04-12"	5

Query 5, subquery:

```
SELECT
    m.member_id,
    m.first_name,
    m.last_name,
    m.member_role
FROM members m
WHERE NOT EXISTS (
    SELECT 1
    FROM attendance a
    WHERE a.member_id = m.member_id
)
ORDER BY m.last_name, m.first_name;
```

OUTPUT:

```
7      "Artyom"  "Amosov" "general"
```

Query 6, club budget balance calculation:

```
SELECT
    SUM(CASE WHEN transaction_type = 'income' THEN amount ELSE 0 END) AS
total_income,
    SUM(CASE WHEN transaction_type = 'expense' THEN amount ELSE 0 END) AS
total_expenses,
    SUM(CASE WHEN transaction_type = 'income' THEN amount ELSE -amount END)
AS current_balance
FROM budget_transactions;
```

OUTPUT:

```
505.00    206.25        298.75
```

REPORT:

This database was created for the purpose of tracking Marist Cybersecurity Club's attendance, membership, and budget to replace the current system with a more consistent and streamlined one. It utilizes four main tables: members, events, attendance, and budget_transactions.

The members table stores member information needed to maintain club communications and track club attendance, which is supported by the events table that holds general event information. The attendance table supports attendance tracking by connecting members to the events they attended. This table is necessary because members and events have a many-to-many relationship, as in one member can attend many events, and one event can have many members attend. The budget_transactions table stores club income and expenses. It can optionally connect a transaction to a specific event through event_id, which allows the club to track event-related spending while still allowing general transactions such as dues, fundraising, or SGA funding to exist without being tied to one event.

This design is normalized to about 3NF by keeping each main subject in its own table and connecting related records with foreign keys where necessary. This helps reduce repeated data and makes the database easier to update, query, and maintain.

The 6 queries performed on the database identify board members, show attendance for a specific event, list event-related expenses, count attendance per event, find members with no attendance records, and calculate the club's current budget balance.

One change from the original project idea was simplifying dues tracking. Instead of creating a separate dues table, dues payments are recorded as income transactions in the budget_transactions table using the dues category. This kept the project focused while still allowing the club to track basic income, expenses, event spending, and budget balance. A more advanced version of the project could add a separate dues table to track which individual members have paid, who still owe dues, and when each payment was submitted. Due to time constraints, I also decided to omit my proposed project extension detailed in the original project proposal.

AI DISCLOSURE:

I used Anthropic's Claude as a tool of support while completing this project. I used it to help interpret the assignment requirements, organize the project structure, and review the database design for feedback.